

# Results

## Study 1: Feasibility of Predicting Keep/Remove Decisions

### Motivation

### Formulating the task

### Dataset

Prolific participants (n=1,321) reviewed pairs of original and mirrored posts (n=959 pairs), resulting in a final dataset of n=13,250 rows, with each row representing a human keep/remove decision for the pair of posts. In these “linked fate” trials, the users saw both the original post and its mirrored equivalent, randomly ordered, and were asked to make a keep/remove decision for the pair. In this dataset, 66.2% of post pairs were kept (and 33.8% removed).

### Training an initial set of models with explainable text features

We first generated a set of explainable text features:

- Does the text involve intergroup discussion?
- Does the text have PRIME content?
- Does the text have a positive valence?
- Does the text have toxic content?
- What is the political stance of the text?
- What is the number of characters in the text?
- What is the average sentence length?
- What is the Flesch-Kincaid reading score?

We then trained two models on these features: a logistic regression model and an XGBoost model. We report those results in a table

Caption: Model performance for predicting keep/remove decisions. Precision, recall, F1, and ROC-AUC are computed with the remove decision as the positive class ( $y = 1$ ).

| Model                          | Split | Accuracy | Precision | Recall | F1    | ROC-AUC |
|--------------------------------|-------|----------|-----------|--------|-------|---------|
| Baseline (always predict keep) | Test  | 0.649    | 0.000     | 0.000  | 0.000 | 0.500   |
| Logistic regression            | Train | 0.699    | 0.572     | 0.407  | 0.475 | 0.675   |
| Logistic regression            | Test  | 0.695    | 0.597     | 0.402  | 0.480 | 0.678   |
| XGBoost                        | Train | 0.737    | 0.667     | 0.430  | 0.523 | 0.777   |
| XGBoost                        | Test  | 0.693    | 0.603     | 0.367  | 0.457 | 0.702   |

### Calibration

In Study 1, ...

(initial results from Study 1)

## **Study 2**

In Study 2, ...

(initial results from Study 2)

### **Model performance**

(model performance)

### **Scaling curves**